

5. Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

```
[11] ✓ 0s
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)
# Normalize the data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Create KNN model
model = KNeighborsClassifier(n_neighbors=3)
# Train the model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Display results
print("Predicted Values:")
print(y_pred)
print("\nActual Values:")
print(y_test)
print("\nAccuracy:", accuracy_score(y_test, y_pred) * 100)
```

OUTPUT:

```
... Predicted Values:
[0 1 1 0 2 1 2 0 0 2 1 0 2 1 1 0 1 1 0 0 1 1 1 0 2 1 0 0 1 2 1 2 1 2 2 0 1
 0 1 2 2 0 1 2 1]

Actual Values:
[0 1 1 0 2 1 2 0 0 2 1 0 2 1 1 0 1 1 0 0 1 1 1 0 2 1 0 0 1 2 1 2 1 2 2 0 1
 0 1 2 2 0 2 2 1]

Accuracy: 97.77777777777777
```